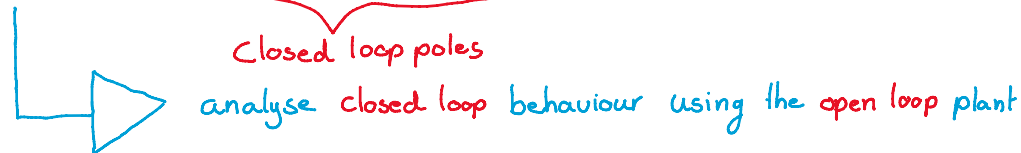


2 - Root Locus design method

Root Locus method

Analyse changes in the system parameters that modify the roots of the characteristic equation



understanding of system

Implication on changes

But first, a side step

$n = \# \text{Poles}$, $m = \# \text{zeros}$

Product operator

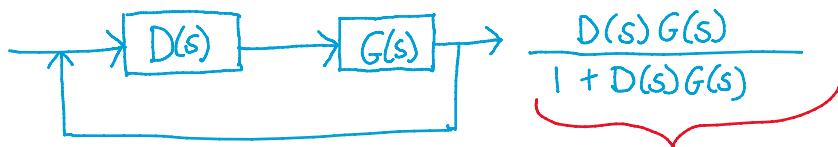
$$G(s) = \frac{s^m + b_1 s^{m-1} + \dots + b_m}{s^n + a_1 s^{n-1} + \dots + a_n} = \frac{(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)} = \frac{\prod_{i=1}^m (s-z_i)}{\prod_{i=1}^n (s-p_i)}$$

Monic: highest power has a coefficient of 1

Strictly proper $n > m$

Bi proper $n = m$

non proper $n < m$



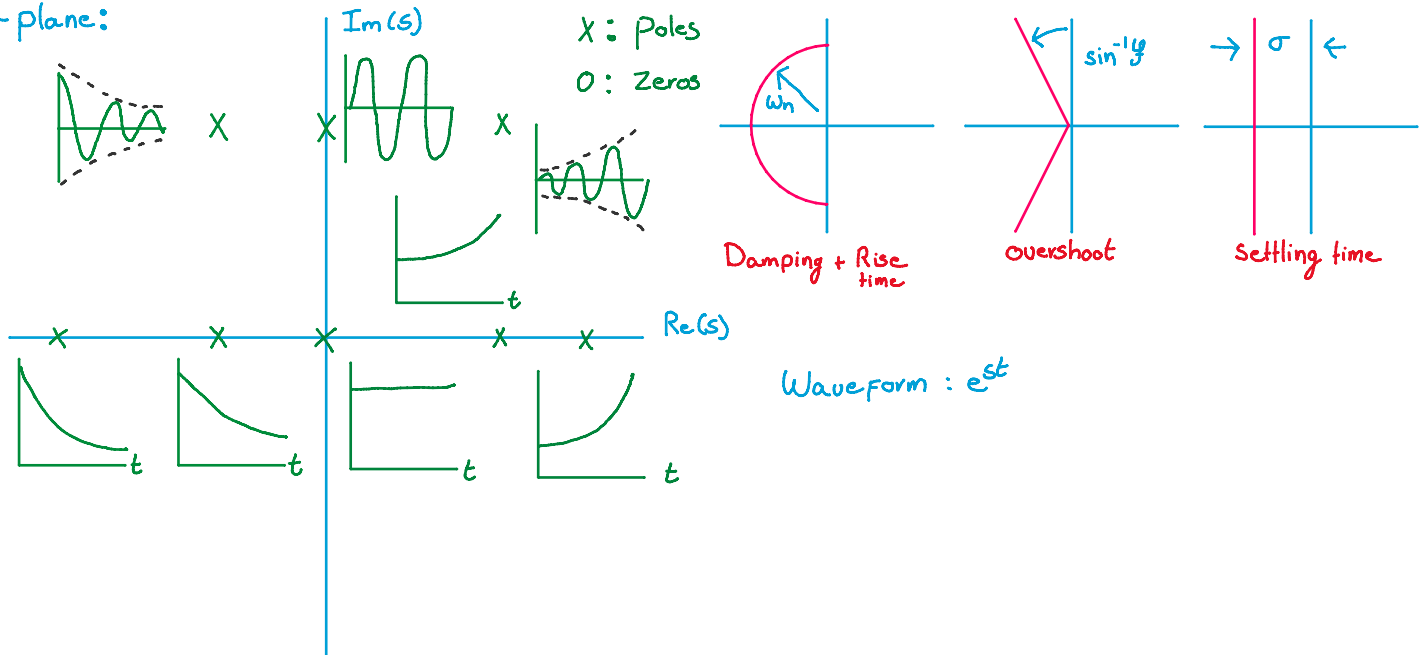
equivalent forms

$$\left. \begin{aligned} 1 + KL(s) &= 0 \\ 1 + K \frac{b(s)}{a(s)} &= 0 \\ a(s) + Kb(s) &= 0 \\ L(s) &= -1/K \end{aligned} \right\}$$

Closed loop system characteristic equation

$$\frac{b(s)}{a(s)} = \frac{\prod_{i=1}^m (s-z_i)}{\prod_{i=1}^n (s-p_i)}$$

S-plane:



Root Locus

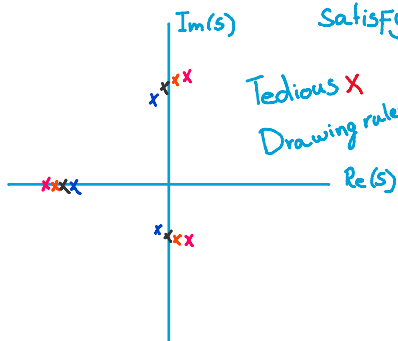
plural: Loci

Locus: position or place where something occurs or is situated.

Curve or figure formed by points
 satisfying an equation

Root Locus: $1 + kL(s) = 0 \Leftrightarrow 1 + D(s)G(s) = 0$
 $k \in [0, \infty)$

open loop plant $\rightarrow$ dynamics closed loop plant



Tedious $\times$
 Drawing rules $\checkmark$

"But my computer can do this"

- Understand change in dynamic response
- Understand consequences on response for controller design

Example:

$$\left. \begin{array}{l} G(s) = \frac{s+2c}{s(s+c)} \\ D(s) = 1 \end{array} \right\} 1 + D(s)G(s) = 1 + \frac{s+2c}{s(s+c)}$$

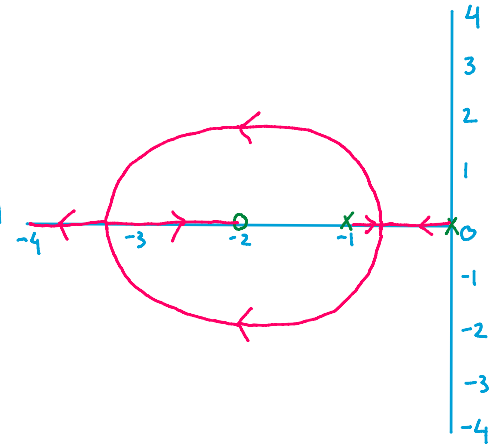
Characteristic equation: $s^2 + s + c(s+2) = 0$

Complex when < 0
 $0.172 < K < 5.828$

$$r_1, r_2 = -\frac{k+1}{2} \pm \frac{\sqrt{k^2 - 6k + 1}}{2}$$

$$\begin{array}{llll} L(s) = \frac{s+2}{s(s+1)} & b(s) = s+2 & m=1 & z_i = -2 \\ a(s) = s(s+1) & n=2 & p_i = 0, -1 & \end{array}$$

$K = c$



Summary

- Closed loop system analysis by varying K and using open loop system
- Solve characteristic equation $1 + KL(s) = 0$
- Changing pole location changes dynamic response
- Demonstration of simple example